evaluator.py

`src/moe_pruning/evaluation/evaluator.py · 215 lines · module moe_pruning.evaluation.evaluator`

What it does

Builds the evaluation examples, runs a model over them, and measures how far each pruned layer's output is from the original layer's output (the reference probe).

How it works

- `prepare_eval_examples()` converts evaluation documents into examples with the task objective (span corruption or next-token) and caches them with a fingerprint of the dataset checksum and settings, so every stage scores exactly the same examples.
- `evaluate_model()` runs batches, checks logits for inf/NaN, and collects per-example loss sums, token counts and teacher-forced argmax predictions through the task.
- `ReferenceExpertProbe` recomputes, for every token, the exact original layer output (original routing, all original experts) and compares it with the pruned layer's real output. Works for bias, drop and reroute and for top-k routing.
- `build_reference_probe()` loads only the removed experts' original weights (a few MB each) from the original checkpoint via a reader, placed on each layer's own device.

Notes

- The probe measures the local error of each layer given its actual (possibly already perturbed) input.

Defined in this file

Extracted from the source code (classes, public functions and methods).

Name	Description
`eval_settings(cfg, task)`	Settings that determine the evaluation examples (dataset checksum + task settings).
`prepare_eval_examples(cfg, tokenizer, paths, task)`	Build the evaluation examples once and cache them; every stage reuses the cache.
`class ReferenceExpertProbe`	Measures ||y_original - y_pruned||^2 of pruned layers on the evaluation data.
`.original_layer_output(layer, moe_input)`	(y_original [B,S,D] float32, original expert indices [B,S,K]).
`.__call__(record, token_mask)`	Add one layer forward record to the error sums.
`.summary()`	Per-layer MSE and relative errors (all tokens / affected tokens).
`build_reference_probe(model, adapter, layers, replaced, reader, device)`	Probe with the original weights of the removed experts (read lazily via `reader`).
`class EvalOutput`	Metrics plus per-example losses, token counts and predictions of one evaluation.
`evaluate_model(model, examples, device, batch_size, pad_token_id, task, adapter, probe, desc)`	Run the model over all examples and collect losses, predictions and probe errors.

Settings it reads

`EVALUATION_DATASET_PATH`, `EVAL_MAX_SAMPLES`, `MODEL_PATH or HF_MODEL_ID`

Connections

Uses: `moe_pruning.config`, `moe_pruning.data.c4_support`, `moe_pruning.evaluation.metrics`, `moe_pruning.model.architecture`,

`moe_pruning.model.instrumentation`, `moe_pruning.model.placement`, `moe_pruning.tasks.base`, `moe_pruning.utils.paths`

Used by: `scripts/evaluate.py`